Home Lab — Technical Documentation

Enterprise IT Infrastructure Build — Complete Reference

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1. Project Overview

This document provides complete technical documentation for the Wizardwerks Enterprise Labs home lab — a multi-server Windows enterprise environment built from scratch to simulate a real-world small business IT infrastructure. The lab was built to demonstrate hands-on enterprise IT administration skills including Active Directory, DNS, DHCP, Group Policy, file services, PowerShell automation, help desk ticketing, patch management, backup and disaster recovery, cloud identity integration, and network security via pfSense.

Lab Name	Wizardwerks Enterprise Labs
Domain Name	CORP.local
Platform	Oracle VM VirtualBox 7.2.6
Host OS	Windows (physical host machine)
Host Storage Available	161 GB free
Build Start	March 2026
Documentation Version	3.0
Administrator	James Mitchell
M365 Tenant	wizardwerks.onmicrosoft.com
GitHub	Public — README, network diagram, tech doc, PowerShell scripts

2. Host Machine

The physical host machine runs VirtualBox and hosts all virtual machines. It serves as the hypervisor layer for the entire lab environment.

Role	Hypervisor — VirtualBox Host
Virtualization Platform	Oracle VM VirtualBox 7.2.6
Free Storage	161 GB available for backup repositories and VM expansion
Network Adapters on VMs	Internal Network (VM-to-VM) + NAT (internet access)
Drag and Drop	Bidirectional — enabled for file transfers between host and VMs
Backup Repository	E: drive on FS01 — dedicated 30GB backup volume

3. Virtual Machines

3.1 — DC01 (Domain Controller)

DC01 is the primary domain controller and the backbone of the entire lab environment. It hosts all core identity and network services.

Machine Name	DC01
Role	Primary Domain Controller
Operating System	Windows Server 2022
Domain	CORP.local
Storage Used	10.39 GB
Storage Available	39 GB
Network Adapter 1	Internal Network — VM-to-VM communication
Network Adapter 2	NAT — internet access for updates and downloads
vCPU	1 vCPU (single core — lab environment)

Services Running on DC01

- Active Directory Domain Services (AD DS)
- DNS Server — authoritative for CORP.local
- DHCP Server — IP address assignment for domain devices
- Group Policy Management
- WSUS (Windows Server Update Services) — patch management, GPO deployed
- Spiceworks Help Desk Agent — cloud help desk integration
- Microsoft Entra ID Connect — hybrid identity sync service, running

Active Directory Configuration

- Forest/Domain: CORP.local
- UPN Suffix: wizardwerks.onmicrosoft.com — added and applied to all users
- Organizational Units: HR, IT, Sales, Corp Users, Workstations
- Security Groups: HR_Users, IT_Users, Sales_Users — one per OU
- Each OU contains test user accounts provisioned via PowerShell
- Group Policy Objects linked to each OU
- Drive mapping GPO configured and verified with gpresult /r

DNS Configuration

- Primary DNS zone: CORP.local — Active Directory integrated
- DC01 is the authoritative DNS server for the internal domain
- All domain devices point to DC01 for DNS resolution
- External DNS forwarding configured for internet name resolution

DHCP Configuration

- DHCP scope configured to assign IPs within defined range
- DC01 serves as DHCP server for all lab VMs
- Lease duration: standard default
- APIPA fallback (169.254.x.x) confirmed as expected behavior when DHCP unavailable

Entra ID Connect Configuration

- AzureADConnect.msi installed on DC01
- Microsoft Azure AD Sync service: RUNNING
- All CORP.local users synced to Entra ID with @wizardwerks.onmicrosoft.com UPNs
- IdFix ran clean — all topleveldomain and blank displayname errors resolved before sync
- SCP configured for Hybrid Entra ID Join via Entra ID Connect wizard
- Manual sync command: Start-ADSyncSyncCycle -PolicyType Delta

3.2 — FS01 (File Server)

FS01 is the dedicated file server hosting department shared drives, home directories, and the backup repository. Windows Server Backup runs scheduled nightly jobs to the dedicated E: backup volume.

Machine Name	FS01
Role	File Server / Backup Server
Operating System	Windows Server 2022
Domain	CORP.local — domain joined
Network Adapter 1	Internal Network — communication with DC01
Network Adapter 2	NAT — internet access confirmed
vCPU	1 vCPU — acceptable for lab environment
Internet Connectivity	Verified — Test-NetConnection google.com = True

File Server Configuration

- Share root: C:\Shares\HomeDirectories
- UNC path: \\FS01\HomeDirectories
- Share permissions: Domain Admins — Full Control, Domain Users — Change
- NTFS permissions: configured per department security group
- Department shares: HR, IT, Sales — each accessible only to corresponding AD security group
- Home directories: created automatically per user via PowerShell provisioning script
- H: drive mapped in AD user profile properties

NTFS Permission Structure

\\FS01\HomeDirectories\HR	HR_Users — Full Control (own folder only)
\\FS01\HomeDirectories\IT	IT_Users — Full Control (own folder only)
\\FS01\HomeDirectories\Sales	Sales_Users — Full Control (own folder only)
All department folders	Domain Admins — Full Control

3.3 — WORKSTATION01

A Windows 10 workstation VM simulates an end user machine within the domain. Used for testing Group Policy application, user login behavior, drive mapping, Hybrid Entra ID Join, and MDM enrollment.

Role	End User Workstation / GPO Testing / Intune Enrollment
OS	Windows 10
Domain	CORP.local — domain joined
Network Adapter 1	Internal Network
Network Adapter 2	NAT — added for internet access
Hybrid Entra ID Join	COMPLETE — AzureAdJoined: YES, DomainJoined: YES (dsregcmd /status verified)
SSO Verified	Clarissa Dane (cdane@wizardwerks.onmicrosoft.com) — M365 portal login with MFA confirmed
MDM Enrollment	BLOCKED — see finding below

MDM Enrollment — Engineering Finding

Hybrid Entra ID Join is complete and verified. MDM auto-enrollment via Intune is blocked by a VirtualBox virtual TPM limitation.

Root cause: Error 0x8018002a — device credential token request fails because VirtualBox's virtual TPM cannot satisfy Azure's hardware-backed device authentication requirement. The GPO (MDM Auto Enrollment), Intune automatic enrollment scope (All users), and user licensing (M365 Business Premium) are all correctly configured. The limitation is at the hypervisor level.

Enterprise context: Production environments using enterprise hypervisors such as Hyper-V or VMware ESXi support hardware TPM passthrough, which resolves this error. Physical hardware with a real TPM chip would complete enrollment successfully. Switching WORKSTATION01 from legacy BIOS to UEFI to enable vTPM was evaluated but deferred — it caused boot failure during prior testing and the risk-to-benefit tradeoff does not justify the disruption in a lab context.

3.4 — pfSense VM (Network Firewall/Router)

pfSense is an open source firewall and router platform running as a dedicated VM. It provides network segmentation, firewall rule management, VLAN capability, and inter-network routing for the Wizardwerks lab environment.

Machine Name	pfSense
Role	Firewall / Router / Network Security
Platform	pfSense CE (Community Edition)
OS Base	FreeBSD
Storage	8 GB — dynamically allocated
Network Adapter 1 (WAN)	NAT — internet facing side
Network Adapter 2 (LAN)	Internal Network — connects to lab environment
LAN IP	192.168.1.1 /24
Status	Installed — initial configuration complete. Firewall rules and VLAN segmentation planned Week 4.

pfSense Current Configuration

- Setup wizard complete: hostname pfsense, domain CORP.local, DNS 192.168.1.10 / 8.8.8.8
- RFC and bogon blocking: ENABLED
- DHCP on LAN: DISABLED — DC01 handles DHCP
- Web GUI accessible: https://192.168.1.1 from Workstation VM
- Admin password changed from default
- Snapshot taken: pfSense — Initial Config Complete

pfSense Planned Configuration

- VLAN segmentation — separate network zones for different lab functions
- Firewall rules — control traffic between VLANs and to/from internet
- Inter-VLAN routing — managed and controlled through pfSense
- RADIUS/VPN — NPS on DC01, OpenVPN on pfSense

4. Cloud and Microsoft 365

4.1 — Microsoft 365 Business Premium

The Wizardwerks lab is integrated with a Microsoft 365 Business Premium trial tenant, enabling cloud identity, Exchange Online, Intune MDM, and Conditional Access.

Tenant	wizardwerks.onmicrosoft.com
Plan	Microsoft 365 Business Premium (trial)
Admin Account	jamesmitchell@wizardwerks.onmicrosoft.com
Exchange Online	Active — all synced users have cloud mailboxes
User Licenses	All synced AD users licensed
SSO Verified	Clarissa Dane — portal.microsoft.com login with MFA from WORKSTATION01

4.2 — Conditional Access

Security Defaults have been disabled in favor of a custom Conditional Access policy providing granular MFA enforcement.

Policy Name	Require MFA All Users
Status	ON — enforced
Scope	All users, All resources
Exclusion	jamesmitchell@wizardwerks.onmicrosoft.com (admin break-glass account)
MFA Method	Microsoft Authenticator app

4.3 — Intune MDM

Automatic Enrollment	ENABLED — MDM user scope: All
WORKSTATION01 Status	Hybrid Entra ID Joined — visible in Entra ID devices
MDM Enrollment	BLOCKED — VirtualBox vTPM limitation (Error 0x8018002a)
Configuration	GPO, licensing, and Intune scope all correctly configured
Resolution	Would complete on physical hardware or enterprise hypervisor (Hyper-V / VMware ESXi)

5. PowerShell Automation

Two PowerShell provisioning scripts were developed and are stored at C:\Scripts\ on DC01. Both scripts automate the full user onboarding workflow including AD account creation, OU placement, security group assignment, home folder creation, NTFS permissions, and H: drive mapping.

5.1 — Single User Provisioning Script

- Interactive Read-Host prompts for all user details
- Yellow confirmation block displays all entered information before proceeding
- Y/N gate — requires explicit confirmation before any changes are made
- Try/Catch error handling throughout
- $CurrentStep variable tracks exactly which step failed if an error occurs
- AD user account creation with full attribute population
- OU placement based on department input
- Security group assignment — adds user to department security group
- Home folder creation at \\FS01\HomeDirectories\[username]
- NTFS permissions set — Full Control for user on their own folder
- H: drive mapping configured in AD user profile
- Cyan success summary displayed on completion

📝 *Script Location: C:\Scripts\NewUser.ps1*

5.2 — Bulk User Provisioning Script (CSV)

- Reads from C:\Scripts\NewUsers.csv
- ForEach loop processes all users in the CSV sequentially
- Same full provisioning workflow as single user script applied per user
- Real-time per-user SUCCESS / FAILED reporting during execution
- Final summary count displayed on completion — Success: X, Failed: X
- Tested with Justice League CSV — 6 users, 6/6 success

📝 *Script Location: C:\Scripts\BulkProvision.ps1*

📝 *CSV Format: FirstName, LastName, Department, Username, Password*

📝 *Test dataset: Batman, Superman, Wonder Woman, Flash, Aquaman, Green Lantern — 6/6 SUCCESS*

6. Group Policy Configuration

6.1 — Drive Mapping GPO

- Linked to: All department OUs — HR, IT, Sales
- Function: automatically maps H: drive to user home directory at login
- Path mapped: \\FS01\HomeDirectories\%username%
- Verified with: gpresult /r — confirmed GPO appearing in Applied Group Policy Objects

6.2 — WSUS GPO

- Specify intranet Microsoft update service location: http://DC01:8530 (both fields)
- Configure Automatic Updates: 4 — Auto download and schedule the install
- Scheduled install: daily at 3:00 AM
- GPO enforced and applied via gpupdate /force — verified

6.3 — BitLocker GPO

- Policy: Allow BitLocker without a compatible TPM — enabled
- Required for VirtualBox environment — no physical TPM chip

6.4 — MDM Auto Enrollment GPO

- GPO Name: MDM Auto Enrollment
- Setting: Enable automatic MDM enrollment using default Azure AD credentials
- Credential type: User Credentials
- Status: Applied — confirmed via gpresult /r on WORKSTATION01
- Enrollment result: BLOCKED by VirtualBox vTPM limitation — see Section 4.3

6.5 — Key GPO Management Commands

- gpresult /r — show all GPOs applying to current user/computer
- gpupdate /force — force immediate Group Policy refresh
- Get-GPO -All — list all GPOs in PowerShell
- gpresult /h C:\report.html — generate HTML Group Policy report

7. Spiceworks Help Desk

7.1 — Configuration

- Platform: Spiceworks Cloud — browser-based, no local server required
- Agent: installed on DC01 for asset and network visibility
- Access: via web browser from any machine on the network
- Account: registered and active

7.2 — Completed Tickets

Ticket #1	John Doe — Unable to access shared drive — Network
Ticket #2	Jane Smith — New user account request — Other
Ticket #3	Tony Stark — Workstation running slow — Hardware
Ticket #4	Bruce Wayne — Password reset — Other

📝 *All 4 tickets completed full lifecycle: Create > Assign > Internal Note > Public Response > Close*

8. Backup & Disaster Recovery

Windows Server Backup is the backup and disaster recovery solution implemented on FS01. It is a native Windows Server role — no additional software required. Backup jobs are scheduled nightly, and restore capability has been verified through a successful test recovery.

8.1 — Configuration

Role	Windows Server Backup — installed via Server Manager
Backup Volume	E: drive — 30GB VDI dynamically allocated, attached to FS01
Schedule	Nightly at 9:00 PM
Backup Type	Full Server
Destination	E: (Backup) volume

8.2 — Restore Verification

- Test restore completed successfully via Windows Server Backup > Recover
- Recovery type: Files and folders
- Restore option: Create copies — preserves existing files
- Result: Completed successfully — backup and restore capability verified

📝 *CRITICAL: A backup without a tested restore is only half the story. Always verify. This has been done.*

8.3 — Tool Selection Rationale

Veeam Backup and Replication was evaluated but caused significant service dependency issues on FS01 and has been fully uninstalled. Windows Server Backup was selected as a native, lightweight, production-grade alternative. This decision reflects real-world judgment — matching the right tool to the environment rather than defaulting to the most complex option.

9. WSUS — Patch Management

Windows Server Update Services (WSUS) provides centralized patch management for the Wizardwerks domain. Installed as a built-in Windows Server role on DC01.

9.1 — Configuration

Upstream Server	Microsoft Update — DC01 pulls directly from Microsoft
Products Synced	Windows Server 2019 and Later (covers Server 2022), Windows 10
Classifications	Critical Updates, Security Updates, Update Rollups
Sync Schedule	Daily automatic synchronization
Content Directory	C:\WSUS

9.2 — Computer Groups

- Domain Controllers — DC01
- Servers — FS01
- Workstations — Windows 10 workstation VM

10. BitLocker Drive Encryption

BitLocker drive encryption was implemented on DC01 to protect data at rest. A Group Policy workaround was used to configure BitLocker without TPM — a supported and documented enterprise configuration.

DC01 Status	ENABLED — C: drive fully encrypted
Encryption Mode	XTS-AES (New Encryption Mode — strongest available)
Unlock Method	Password
Recovery Key	Saved securely

11. Network Architecture

Subnet	192.168.1.0/24
DC01	192.168.1.10 (static)
FS01	192.168.1.30 (static)
pfSense LAN	192.168.1.1 (gateway)
WORKSTATION01	DHCP from DC01

Connectivity Verification Commands

- Test-NetConnection -ComputerName google.com -Port 80 # Test internet
- Test-NetConnection -ComputerName DC01 -Port 389 # Test AD connectivity
- Test-NetConnection -ComputerName FS01 -Port 445 # Test file share
- nslookup CORP.local # Test DNS resolution

12. Key Commands Reference

Active Directory

- Get-ADUser -Identity username # Get user details
- Get-ADGroupMember -Identity GroupName # List group members
- Search-ADAccount -LockedOut # Find locked accounts
- Unlock-ADAccount -Identity jdoe # Unlock account
- New-ADUser -Name "John Doe" ... # Create user
- Add-ADGroupMember -Identity Group -Members u # Add to group

DNS and Network

- ipconfig /all # Full network config
- ipconfig /flushdns # Clear DNS cache
- ping 8.8.8.8 # Test internet routing
- nslookup hostname # DNS lookup
- tracert hostname # Trace network path

Group Policy

- gpupdate /force # Force immediate GP refresh
- gpresult /r # Show applied GPOs
- gpresult /h C:\report.html # Generate HTML GP report

Entra ID Connect

- Start-ADSyncSyncCycle -PolicyType Delta # Force delta sync
- Start-ADSyncSyncCycle -PolicyType Initial # Force full sync
- Get-ADSyncScheduler # Check sync schedule status

Domain Controller Health

- dcdiag /test:all # Full DC diagnostic
- netdom query fsmo # Show FSMO role holders
- repadmin /showrepl # Show AD replication status

File and Share

- net use H: \\FS01\HR # Manual drive map
- net share # List all shares
- icacls C:\Shares\HR # View NTFS permissions

BitLocker

- manage-bde -status C: # Check BitLocker status
- manage-bde -on C: # Enable BitLocker on C: drive
- Get-BitLockerVolume # PowerShell status check

13. Complete Lab Status Summary

Component	Status	Notes
VirtualBox Hypervisor	**Complete**	Host machine running, all VMs operational
DC01 — Windows Server 2022	**Complete**	Primary domain controller, fully configured
FS01 — Windows Server 2022	**Complete**	File server, domain joined, shares configured
WORKSTATION01 — Windows 10	**Complete**	Domain joined, Hybrid Entra ID joined, SSO verified
pfSense Firewall/Router	**In Progress**	Installed, initial config complete. Firewall rules and VLANs
Active Directory	**Complete**	CORP.local domain, OUs, security groups, users
DNS	**Complete**	Authoritative for CORP.local, forwarding configured
DHCP	**Complete**	Scope configured, IP assignment working
Group Policy	**Complete**	Drive mapping, WSUS, BitLocker, MDM GPOs deployed
NTFS Permissions	**Complete**	Dept shares with RBAC security group access
Home Directories	**Complete**	\\FS01\HomeDirectories, H: drive mapped
BitLocker — DC01	**Complete**	C: drive encrypted, XTS-AES, GPO workaround, verified
PowerShell — Single User	**Complete**	Interactive provisioning with confirmation gate
PowerShell — Bulk CSV	**Complete**	6/6 Justice League test — all success
Spiceworks Help Desk	**Complete**	Cloud platform, agent on DC01, 4 tickets resolved
Windows Server Backup	**Complete**	Scheduled nightly jobs, verified restore, E: volume
WSUS	**Complete**	Installed on DC01, GPO deployed, daily sync configured
Entra ID Connect	**Complete**	Hybrid identity sync running, all users synced to Entra ID
M365 Business Premium	**Complete**	Tenant active, Exchange Online, all users licensed
Conditional Access / MFA	**Complete**	Policy active, MFA enforced all users, SSO verified
Hybrid Entra ID Join	**Complete**	WORKSTATION01 — AzureAdJoined: YES, DomainJoined: YES
Intune MDM Enrollment	**BLOCKED**	vTPM limitation in VirtualBox — Error 0x8018002a. Resolves on physical hardware or enterprise hypervisor.
Intune Compliance Policies	**Planned**	Encryption, password, update status requirements
CIS Hardening	**Planned**	Week 2 — security baselines on DC01 and FS01

SQL Server Express	**Planned**	Install on FS01, SSMS on workstation
Kali Linux VM	**Planned**	Nmap, OpenVAS, simulated attack and defense
Microsoft Sentinel	**Planned**	M365 data connectors, attack detection
RADIUS / VPN	**Planned**	NPS on DC01, OpenVPN on pfSense
pfSense VLANs / Rules	**Planned**	Full network segmentation
Splunk Free	**Planned**	Log ingestion from DC01/FS01, dashboards
ServiceNow PDI	**Planned**	ITSM workflows, linked to lab